

An Efficient Blockchain-Based Conditional Privacy-Preserving Authentication Protocol for VANETs

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Abstract—Vehicular Ad-hoc Networks (VANETs) have potential applications in improving the efficiency and safety for intelligent transportation systems. The openness of VANETs, however, also introduces privacy and security implications. Despite a number of conditional privacy-preserving authentication (CPPA) schemes with anonymity and conditional traceability have been designed for VANETs, a majority of these schemes cannot be directly applied to a real-world setting (e.g., due to the need for a certificate manager for issuing keys in PKI-based solutions, or the inherent key escrow problem in ID-based solutions). There have also been attempts to design blockchain-based CPPA schemes, but these schemes may not support key revocation or are inefficient (e.g., due to on-chain operations). This paper proposes an efficient blockchain-based CPPA (EBCPPA) scheme, which is designed to mitigate the above limitations. Our proposal consists of two key building blocks, namely: signature of knowledge and smart contract. To evaluate the feasibility, we present the security and performance analyses of EBCPPA. Specifically, the performance evaluations show that EBCPPA is

more efficient than other existing state-of-the-art solutions, in terms of signing (improving at least 49.71%), the verification (improving at least 32.84%) and bandwidth requirement (reducing at least 27.59%).

Index Terms—Blockchain, VANETs, Conditional privacy-preserving, Signature of knowledge, Smart contract.

I. INTRODUCTION

AS FIFTH-generation (5G) communication and Internet of Things (IoT) become commonplace, the role of vehicular ad-hoc network (VANET) in intelligent transportation systems will also become more pronounced. For example, due to its highly mobile and rapidly changing network topology, VANETs allow vehicles to exchange real-time traffic data with nearby vehicles, roadside units (RSUs) as well as other supporting infrastructures. Such information sharing can enhance traffic efficiency and minimize the risk of traffic incidents. A typical VANET contains six entities, namely: Traffic control center (TCC), Key generation center (KGC), Internet, RSU, Vehicle and On-board unit (OBU) – see Fig. 1. In the simplified figure, the data flow consists of vehicle-to-vehicle (V2V) and the vehicle-to-infrastructure (V2I) communications. In order to achieve short-range wireless communication, an OBU is installed in each vehicle to share data with other entities (e.g., vehicles or RSUs). The TCC collects these traffic-related messages via RSUs and provides real-time traffic services, such as re-routing, monitoring and managing traffic lights [1].

Security and privacy are the key issues in practical applications of VANETs. Due to the openness and mobility features, VANETs will suffer from message interception, sniff, modification, replay, or a broad range of other attacks (e.g., advanced persistent threats (APTs)). For example, without authentication and privacy protection, an attacker can easily obtain private information (e.g., location, historical route, or license plate information) or inject fabricated traffic information (e.g., status). In extreme case, these malicious behaviors may result in fatal traffic accidents. Thus, the authenticity, anonymity and integrity of transmitted messages should be guaranteed in VANETs.

While message anonymous authentication [2], [3] is a potential solution, absolute anonymity may lead to the spreading of maliciously crafted messages (e.g., information and commands). Thus the lack of tracing is another hindrance to the broader

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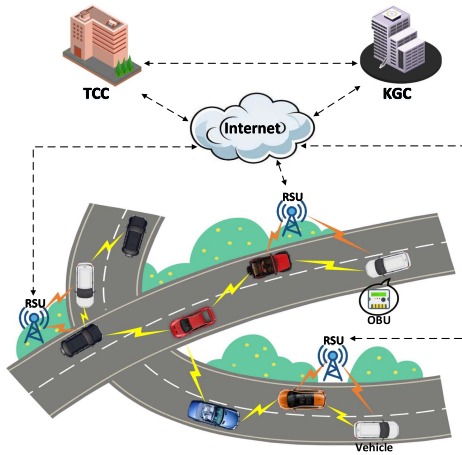


Fig. 1. A typical structure of VANET.

deployment of VANETs. Not surprisingly, a number of conditional privacy-preserving authentication (CPPA) schemes have been proposed to alleviate the above problem. These schemes not only ensure the identity authentication and content integrity, but also achieve conditional privacy protection. That is, during information sharing, the identity of the vehicle is not revealed to any RSUs and vehicles, but it is traceable for a trusted third-party (referred to as the manager in the rest of this paper) if the need arises (e.g., in an investigation or the presentation of a court order).

Existing CPPA protocols for VANETs can be broadly classified into public key infrastructure (PKI)-based (e.g., [4], [5], [6]) and identity (ID)-based schemes (e.g., [7], [8], [9]). The former approaches can be deployed and maintained easily, but generally incur high storage costs (e.g., the need to store potentially number of temporary certificates in the OBUs to guarantee anonymity, particularly as the system scales up) and complex certificate management issues. Although ID-based solutions can mitigate these issues associated with PKI-based solutions and also improve efficiency, they still suffer from key escrow. That is, vehicles' secret keys will be leaked once the key generation center (KGC) is compromised. In addition, we observe that PKI-based and ID-based schemes either rely on a centralized server or do not scale for large number of participants.

In recent years, there have been efforts to integrate blockchain in the construction of CPPA protocols. Blockchain is a distributed ledger and composed by different technologies, mainly cryptography, distributed system and database. The privacy-preserving application can utilize the features of blockchain, such as verifiability, immutability and pseudo-anonymity, to achieve a trade-off in security, privacy and efficiency. In addition, the smart contract deployed in blockchain can also be designed to support autonomous execution with programmed logic. Hence, it is possible to employ blockchain to automatically authenticate message, monitor behavior and trace malicious users [10], [11].

Although existing blockchain-based CPPA (BCPPA) protocols are striving to achieve both anonymity and conditional traceability, they are generally inefficient in a VANET setting. That is, these BCPPA schemes for VANETs involve frequent

interactions and incur high computation cost. Hence, Lin et al. [12] and Feng et al. [13] proposed two BCPPA schemes with anonymity and traceability, respectively, but both schemes are still inefficient in message verification and identity tracking. Therefore, we are motivated to design a more efficient BCPPA scheme with enhanced security for VANETs.

Specifically, to achieve improved authentication efficiency and limit tracing privilege in VANETs, we introduce an efficient blockchain-based conditional privacy-preserving authentication protocol (EBCPPA). A summary of our approach in this paper is as detailed:

- We first design a new structure of the anonymous public key, which we use as a building block for our proposed signature of knowledge (SoK) scheme to achieve authentication. Due to the traceability of anonymous public keys, our EBCPPA ensures that participants authenticate each other anonymously and only the trusted third party (i.e., TA) can extract the real identity. In addition, no expensive secure channel is needed to distribute secret keys in our solution.
- Then, we implement EBCPPA based on the Ethereum test blockchain and VANETs simulator (i.e., VanetMobSim and NS-2), as well as showing the advantages of EBCPPA in comparison to two other recent related schemes, for message signing, verification and identity tracking. Additionally, we perform a security analysis to illustrate that our proposal meets the security and privacy requirements in VANETs.

The remainder of this paper is organized as follows. Section II briefly introduces other existing CPPA schemes designed to operate in a VANET environment, prior to explaining our proposal in Section III. Section IV introduces our system building blocks, including SoK protocol and smart contract. We present our EBCPPA scheme in Section V, followed by its security analysis and performance evaluation in Sections VI and VII, respectively. This paper is summarized in the final section.

II. RELATED WORK

To achieve authentication, anonymity and traceability for VANETs, a large number of CPPA protocols have been proposed by the research community. These schemes can be broadly classified into PKI-based, ID-based and blockchain-based schemes.

In 2007, Raya et al. [14] used the modified PKI infrastructure to satisfy secure authentication requirements for VANETs. Their scheme used an anonymous certificate (AC) to protect privacy. However, there is an unsolved issue that: numerous certificates had to be preloaded and exchanged among RSUs and vehicles.

To deal with the above issue, Lu et al. [15] introduced an RSU-based CPPA protocol with temporary AC. In their proposal, the vehicle's OBU could obtain temporary AC frequently from nearby RSU. While their scheme provides conditional privacy protection, the data sharing procedure requires the CA to manage a large number of certificates. After that, other PKI-based CPPA literature were also proposed [8], [16], [17], but most of them suffered from similar weaknesses as mentioned above.

As another orthogonal line of work, the ID-based CPPA is proposed for figuring out certificate management and exchange problems. Zhang et al. [18] designed an ID-based CPPA scheme, which adopted an aggregate signature function to achieve the low verification cost. However, Horng et al. [19] identified that Zhang et al.'s scheme [18] cannot withstand impersonation attack.

Subsequently, He et al. [7] and Li et al. [20] designed two secure ID-based CPPA schemes, which can guarantee the validity of messages and the conditional privacy of senders simultaneously. While ID-based solutions have advantages in utilization and management compared with PKI-based schemes, they suffer from the key escrow problem and generally require a secure channel.

Due to the features of decentralization, immutability and anonymity [21], blockchain becomes a potential technology to satisfy the functionality and security requirements in VANETs [22], [23], [24], [25]. In 2017, Rowan et al. [26] proposed a blockchain-based PKI framework for V2V secure communication. However, their scheme is vulnerable to malicious attacks. Then, Lu et al. [27] leveraged blockchain to design a privacy-preserving authentication scheme for VANETs. Their scheme provided efficient certificate revocation, but the vehicles need to frequently interact with a certificate authority (CA) to obtain anonymous certificates.

In 2020, Gabay et al. [28] formulated and constructed a blockchain-based anonymous authentication scheme, which integrated Pederson commitment and the token-based mechanism. Although their scheme provided a strong privacy strategy, it ignored the traceability requirement of trusted authority. Taking conditional privacy protection into account, Fen et al. [13] integrated blockchain with attribute-based encryption, but the efficiency of authentication and traceability is still far from application.

Recently, Lin et al. [12] introduced a key derivation function to design the BCPPA system, which cuts down the number of secret keys stored in vehicles. However, Lin et al. [12]'s scheme exists a linear relation between the searching amount of public keys and the cost of tracing. He et al. [29] proposed a hierarchical blockchain-assisted CPPA scheme which can achieve unlinkability, but it is inefficient. Thus, we are motivated to construct an efficient BCPPA scheme in VANETs especially further improving the verification and tracing efficiency.

III. PRELIMINARY KNOWLEDGE

In this section, we first list the relevant notations. Then, we give the system architecture of VANETs and basic definitions of cryptographic primitives, including ElGamal encryption, NIZK argument, signatures of knowledge and security requirements.

A. Notations

For convenience, the notations involved in this paper are shown in Table I.

TABLE I
INVOLVED NOTATIONS

Notation	Description
V_i	The i_{th} vehicle
F_q	A prime finite field
E/F_q	An elliptic curve over F_q
$\mathbb{G}$	An additive cyclic group
q	An order of $\mathbb{G}$
P	A generator of the group G
H_i	One-way collision-resistant hash functions
RID_i	The real identity of entity i
MPK	The public key of TA
msk	The private key of TA
PK_i	The public key of entity i
sk_i	The private key of entity i
t_i	The system timestamp
cid	The smart contract identity
sid_i^j	The i -th entity's j -th anonymous key and traceable information index.
$C_i^{1,k}, C_i^{2,k}$	The i -th entity's j -th anonymous public key
$C_i^{1,k}, C_i^{3,k}$	The i -th entity's j -th traceable information
σ_i	The signature of the message

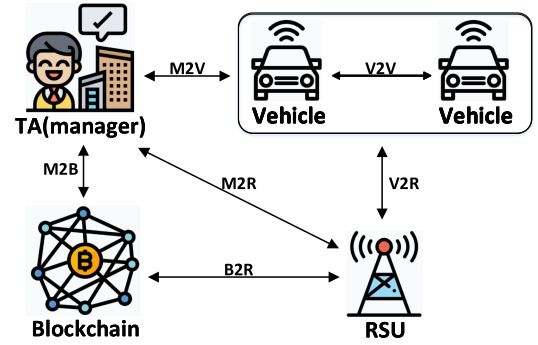


Fig. 2. System architecture of EBCPPA.

B. System Architecture

This section introduces the VANETs system architecture used in this paper, involving **Trusted Authority (TA)**, **RSU**, **Vehicle** and **Blockchain** (see Fig. 2). In our system, each vehicle generates key pair independently by OBU, and thus we do not need KGC to maintain certificate or generate key pair. In addition, **TA**, whose capacity is the same as **TCC**, is the trusted system manager and auditor.

Our proposed system is established with six methods of communication, namely M2V, M2B, M2R, V2V, V2R and B2R, where M2V is for vehicles to register their identities with the TA manager, M2B is for TA to submit anonymous public key and other information into blockchain, M2R is for TA to communicate with RSUs, V2V is for vehicles to wirelessly exchange messages among them, V2R is the wireless communication between vehicles and RSUs to share traffic-related data and blockchain's transactions, and B2R is for RSUs to query information from the blockchain.

- **TA:** TA is the intelligent transport service provider, also the manager of blockchain's smart contract. This entity is with

enough memory and computation capability, and trusted to all participants in VANETs. TA accounts for generating anonymous public keys for all users and uploading key materials via the smart contract. Moreover, It can trace users' real identities and revoke user privilege (if necessary).

- **Vehicle:** The vehicle is equipped with various sensors and can wirelessly broadcast collected data to other vehicles and RSUs. The vehicle has limited storage space and computation ability. Besides, it uses OBU (with the tamper-proof property) to store private keys and other sensitive information.
- **RSU:** This entity, located on the roadside, acts as a gateway between vehicles and service infrastructures (e.g., TA, Blockchain). This wireless communication unit can achieve connectivity to surrounding vehicles and verify the validity of received messages.
- **Blockchain:** The Blockchain provides immutable data storage and automated smart contract. This network used in our scheme is responsible for storing and managing anonymous public keys in the smart contract.

C. Blockchain

As a decentralized system, blockchain is characterized by its decentralization, transparency, automation and tamper-resistant. Distributed nodes jointly maintain a ledger via a consensus mechanism and the result can be viewed by all participants. There are two main types of blockchain, including the permissionless blockchain (e.g., Bitcoin and Ethereum) and the permissioned blockchain (e.g., Hyperledger). The former type allows anyone to maintain the blockchain, while the latter one only allows the authenticated nodes to take part in consensus. In our work, we deploy transactions and smart contract of our scheme in Ethereum.

- **Transaction:** The transaction mainly consists of the addresses of sender/receiver, the transferred value and a signature. The block stores a number of transactions in the Merkle tree data structure. Each block calculates the cryptographic hash of the prior block in the blockchain, and then links the blocks together. This structure achieves the tamper-proof of on-chain data. To establish reliability and trust in blockchain, consensus mechanism (e.g., PBFT) achieves that the participators reach necessary agreement about the present state.
- **Smart Contract:** Smart contracts are written in a Turing complete programming language (e.g., Solidity, Vyper, DAML) and stored on the blockchain. They autonomously execute in a digital form and can achieve computerized transaction protocols from simple to more complex functions. Each smart contract with a unique public address can be triggered internally and externally. In our scheme, the smart contract is mainly responsible for public key information storing, viewing, and deleting according to the caller's permissions.

D. ElGamal Encryption

Our proposal uses the asymmetric ElGamal encryption scheme [30] which can be proven secure in the IND-CPA model. This scheme involves three algorithms, including Key generation (**PKG**), Encryption (**Enc**) and Decryption (**Dec**).

- $(sk, PK) \leftarrow \mathbf{PKG}(\lambda)$: This algorithm chooses a group G with prime order q and a generator g . Then, selecting a number $sk \in \mathbb{Z}_q^*$ randomly as the private key, and calculating $PK = g^{sk}$ as the public key.
- $(C_1, C_2) \leftarrow \mathbf{Enc}(PK, m)$: This algorithm selects a random number $u \in \mathbb{Z}_q^*$ and calculates $C_1 = g^u$, $C_2 = m + PK^u$. Finally, it outputs (C_1, C_2) as ciphertext.
- $m \leftarrow \mathbf{Dec}(sk, C_1, C_2)$: The message will be decrypted by computing $m = C_2 - C_1^{sk}$.

E. NIZK Argument

A non-interactive zero-knowledge (NIZK) argument [31] includes a prover, a verifier and a message transmitted between them. The verifier can convince the prover of knowing certain values without obtaining any other useful information.

Suppose $R = \{x, w\}$ is a computable relation, where x is a public instance and w is a witness. Here, the prover calculates a verifiable proof π using x and w . When receiving a single proof π from the prover, the verifier can check it upon x .

There involved three algorithms (G, P, V) in NIZK argument: Common reference string (CRS) generation $G_{crs}(\lambda)$, Proof $P(crs, x, w)$ and Verify $V(crs, x, \pi)$. With relation R , a protocol is said to be an NIZK argument if it satisfies the following properties:

- 1) **Completeness:** For $\forall crs \leftarrow G_{crs}(\lambda)$ and $\forall (x, w) \leftarrow R$, where crs is a common reference string, we have

$$\Pr \left[crs \leftarrow G_{crs}(\lambda), (x, \pi) \leftarrow \mathcal{A}(crs) : \begin{matrix} V(crs, x, \pi) = 1 \end{matrix} \right] = 1 - \text{negl}(\lambda).$$

- 2) **Soundness:** For all $\mathcal{PPT}$ adversaries $\mathcal{A}$, we have

$$\Pr \left[crs \leftarrow G_{crs}(\lambda), (x, \pi) \leftarrow \mathcal{A}(crs) : \begin{matrix} x \notin \mathcal{L} \wedge V(crs, x, \pi) = 1 \end{matrix} \right] = \text{negl}(\lambda).$$

- 3) **Zero-Knowledge:** There are two $\mathcal{PPT}$ simulators $\mathcal{S} = (\mathcal{S}_1, \mathcal{S}_2)$ and a trapdoor information td such that.

$$\begin{aligned} & \Pr \left[\begin{matrix} crs \leftarrow G_{crs}(\lambda) \\ (x, w) \leftarrow \mathcal{A}_1(crs) : (x, w) \in R \wedge \\ \pi \leftarrow P(crs, x, w) : \mathcal{A}_2(crs, x, \pi) = 1 \end{matrix} \right] \\ & - \Pr \left[\begin{matrix} (crs, td) \leftarrow \mathcal{S}_1(\lambda) \\ (x, w) \leftarrow \mathcal{A}_1(crs) : (x, w) \in R \wedge \\ \pi \leftarrow \mathcal{S}_2(crs, x, td) : \mathcal{A}_2(crs, x, \pi) = 1 \end{matrix} \right] \\ & = \text{negl}(\lambda). \end{aligned}$$

F. Signatures of Knowledge

Signatures of knowledge (SoK) [32] is a non-interactive proof system that allows one entity to convince other parties of knowing certain values without revealing them. Also, the SoK owns the functionality of a digital signature. There are three algorithms in the SoK, including Setup, GenProof and VerfProof.

- $SP \leftarrow \text{Setup}(1^\lambda)$: This algorithm generates system parameters with λ as input, and returns system parameters SP .
- $\pi \leftarrow \text{GenProof}(SP, m, x, w)$: This algorithm generates SoK proof with system parameters SP , message m and hard relation $(x, w) \in R$ (x is the public statement, w is the secret witness) as inputs, and returns a proof π .
- $\{0, 1\} \leftarrow \text{VerfProof}(SP, m, x, \pi)$: This verification algorithm takes system parameters SP , message m , statement x and SoK proof π as inputs, and returns 0 if π is invalid or 1 otherwise.

G. Security Requirements

Considering a variety of attacks in the real world, a practical BCPPA scheme for VANETs should satisfy the following essential security requirements [7], [33].

- 1) **Message authentication**: The vehicle and the RSU could validate received messages from a registered communicator, including authenticity, integrity and freshness. Moreover, any tampering of the message will be detected.
- 2) **Identity anonymity**: During authentication, the scheme should guarantee the user's identity privacy, such that no malicious adversary can extract the real identity of the message sender.
- 3) **Non-repudiation**: No message signer can deny the authenticity of signature on the system. This can prevent attackers from denying or covering up illegal behaviors.
- 4) **Conditional traceability**: Conditional traceability stipulates that only the trusted third party can trace the identity of the malicious or misbehavior user. Other entities do not have the qualification to identify participants. Therefore, the authority can take necessary regulatory actions (e.g., law enforcement) on the malicious users.
- 5) **Un-linkability**: To avoid attackers to trace the vehicle's historical path, messages sent by the same vehicle are unable to be linked.
- 6) **Resistant against other common attacks**: The blockchain-based CPPA scheme should also resist various common attacks in VANETs (e.g., replay attack, modification attack, impersonation attack).

IV. PROPOSED SYSTEM BUILDING BLOCK

In this section, we design two types of system building blocks, including the SoK scheme and the smart contract. They are the essential parts of our proposed system.

A. Designed Signature of Knowledge

We construct a new signature of knowledge (SoK) scheme for vehicles and RSUs to achieve authentication in communications. In our EBCPPA, each vehicle owns private key (sk_i), public key ($PK_i = sk_i P$) and anonymous public keys ($C_i^{1,j} = u_j P$, $C_i^{2,j} = u_j PK_i + PK_i$, where $u_j \in Z_q^*$). On basis of our SoK scheme, the vehicle can prove to a verifier (e.g., RSU or vehicle) that it owns the private key without revealing it. In addition, our

SoK scheme also serves as a signature to ensure the authentication and legitimacy of transmitted messages.

We define the above SoK scheme as $SoK\{sk_i : C_i^{2,j} = sk_i C_i^{1,j} + sk_i P\}(msg_i, t_i)$, where sk_i is the private key, $\{C_i^{1,j}, C_i^{2,j}\}$ is the anonymous public key, msg_i is the message to be signed and t_i is the timestamp. Our SoK scheme consists of three algorithms: **GenProof**, **VerifyProof** and **BVerifyProof**. The brief description is shown below.

- $(msg_i, \sigma_i, t_i) \leftarrow \text{GenProof}(sk_i, x_i, msg_i, t_i)$. This proof generation algorithm takes a witness sk_i , public information $x_i = (C_i^{1,j}, C_i^{2,j})$, a message msg_i and a timestamp t_i as an input. It first selects $r_i \in Z_q^*$ randomly and calculates $R_i = r_i C_i^{1,j} + r_i P$, $e_i = H(P, C_i^{1,j}, C_i^{2,j}, R_i, msg_i, t_i)$. Then, it computes $z_i = r_i - e_i sk_i \pmod{q}$. Finally, it sets $\sigma_i = \{z_i, R_i\}$ and outputs message/signature pair $\{msg_i, \sigma_i, t_i\}$.
- $\{0, 1\} \leftarrow \text{VerifyProof}(x_i, msg_i, \sigma_i, t_i)$. This proof verification algorithm takes public information $x_i = (C_i^{1,j}, C_i^{2,j})$, a signature σ_i , a message msg_i and a timestamp t_i as an input. It first parses $\sigma_i = \{z_i, R_i\}$. Then, it compute $e_i = H(P, C_i^{1,j}, C_i^{2,j}, R_i, msg_i, t_i)$ and checks whether the equation $R_i = z_i(C_i^{1,j} + P) + e_i C_i^{2,j}$ holds. If it holds, the algorithm outputs 1 to represent the validity of the message/signature pair, and 0 otherwise.
- $\{0, 1\} \leftarrow \text{BVerifyProof}(\{x_i, msg_i, \sigma_i, t_i\}_{i=1}^n)$. The batch verification algorithm takes multiple message/signature pairs $\{x_i, msg_i, \sigma_i, t_i\}_{i=1}^n$ as an input. It first parses $x_i = (C_i^{1,j}, C_i^{2,j})$ and $\sigma_i = \{z_i, R_i\}$, where $i \in \{1, 2, \dots, n\}$, respectively. Then, it selects a vector $v = v_1, v_2, \dots, v_n$, where $v_j \in [1, 2^t]$, $j \in \{1, 2, \dots, n\}$ and t is a small integer. After that, it calculates $e_i = H(P, C_i^{1,j}, C_i^{2,j}, R_i, msg_i, t_i)$. Finally, it checks whether $\sum_{i=1}^n v_i R_i = \sum_{i=1}^n v_i z_i C_i^{1,j} + (\sum_{i=1}^n v_i z_i) P + \sum_{i=1}^n v_i e_i C_i^{2,j}$ holds or not. If it holds, the algorithm outputs 1 to represent the validity of message/signature pairs, and 0 otherwise.

We utilize the small exponent test technology to achieve the non-repudiation of signatures in **BVerifyProof**. A vector v , consisting of small random integers v_i , is used to quickly detect any modification of a batch of message/signature pairs. If there is no randomness, an adversary can easily generate invalid message/signature pairs but acceptable by the verifier.

B. Design of Smart Contract

We design a smart contract (see Algorithm 1) to achieve anonymous public key management. It can be invoked by TA (the smart contract's manager), message senders and verifiers to execute corresponding operations in the smart contract. During the vehicle registration, the vehicle transmits its real identity and public key to TA (the smart contract manager). Only TA can generate and submit the corresponding anonymous public key of the vehicle into the smart contract. During authentication, the verifier can retrieve target vehicle's anonymous public key to check the validity of messages. In order to avoid fake or faulty

Algorithm 1: Smart Contract on PKList.

Require: Function name, invoked parameters
Ensure: *Setting up functions:*
 address manager; % Initialize the issuer (TA) as the manager

function APKList()
 % Constructor, automatically invokes when this smart contract is deployed.
 manager = msg.sender; % Define the deployer as the TA
 struct APK
 {
 C_1 ; % uint256[2], fist part of an APK
 C_2 ; % uint256[2], second part of an APK
 C_3 ; % uint256[2], third part of an APK
 }
 mapping(uint256 $\Rightarrow$ APK) apk % A mapping of APK

function Submit(uint256 cid, uint256[2] C_1 , uint256[2] C_2 , uint256[2] C_3) public returns (address addr)
 % Invoked by TA to add a new anonymous public key.
 require(msg.sender == manager); % Only TA can successfully execute this algorithm
 apk[cid]. C_1 = C_1 ;
 apk[cid]. C_2 = C_2 ;
 apk[cid]. C_3 = C_3 ;
return msg.sender;

function Get(cid) view returns (uint256[2] C_1 , uint256[2] C_2)
 % Invoked by any to retrieve the anonymous public key.
return apk[cid]. C_1 , apk[cid]. C_2 ;

function Trace(cid) view returns (uint256[2] C_1 , uint256[2] C_3)
 require(msg.sender == manager); % Only TA can successfully execute this function
return apk[cid]. C_1 , apk[cid]. C_3 ;

function Revoke(uint256 cid) public returns (bool)
 % Invoked by the manager to revoke an APK.
 require(msg.sender == manager); % Only TA can successfully execute the revocation
 apk[cid]. C_1 = NULL;
 apk[cid]. C_2 = NULL;
 apk[cid]. C_3 = NULL;
return 1;

return 0;

message spreading, TA can trace real identity of malicious users and revoke their privileges.

V. PROPOSED ARCHITECTURE

We will present the construction of our proposed scheme based on our designed system building block. The scheme mainly includes six phases: **System Setup**, **Smart Contract Deployment**, **Registration**, **Message Signing**, **Message Verification** and **Track and Revocation**.

A. System Setup

TA initializes the smart contract as the manager. Then, vehicle can interact with TA to register and generate anonymous public keys. TA and the vehicle carried out the following steps.

- 1) **Basic initialization:** TA selects a cyclic additive group $(\mathbb{G}, q, P)$ and chooses a one-way hash function $H: \{0, 1\}^* \rightarrow \{0, 1\}^*$. Then, TA generates a random number $msk \in \mathbb{Z}_q^*$ as its private key and calculates the public key $MPK = mskP$. Finally, TA publishes system parameters $\{\mathbb{G}, q, P, H, MPK\}$.
- 2) **Blockchain initialization:** TA maintains a configuration file to establish a blockchain. For simplicity, TA can also directly join an existing blockchain system and uses its capacity.

B. Smart Contract Deployment

In this phase, TA deploys the smart contract *PKList* (see Algorithm 1) into the blockchain. The identity of this smart contract *scid* is published for all participants to invoke it. By utilizing smart contract's functions, the system can provide anonymous public key management for vehicles and RSUs. In addition, it supports key revocation when TA detects malicious users or users unsubscribe proactively.

C. Registration

Before participating in data interaction, every vehicle and RSU need to register real identity in the controlled way. Detailed steps of registration are shown as follows.

- 1) The vehicle V_i selects a random number $sk_i \in \mathbb{Z}_q^*$ as its private key and calculates its corresponding public key $PK_i = sk_iP$. Then, V_i sends its real identity RID_i and the public key PK_i to TA via a secure channel.
- 2) On basis of the ElGamal encryption, TA uses PK_i and MPK to encrypt V_i 's public key, respectively. That is, TA selects a random number $u_j \in \mathbb{Z}_q^*$ to compute $C_i^{1,j} = u_jP$, $C_i^{2,j} = u_jPK_i + PK_i$. TA computes $C_i^{3,j} = u_jMPK + PK_i$. Then, TA computes the index $cid_i^j = H(C_i^{1,j} || C_i^{2,j})$.
- 3) TA keeps the vehicle's real identity, the public key and the key index secretly. Then, it sends cid_i^j to V_i .
- 4) Finally, TA invokes Algorithm 1 (i.e., Submit($cid_i^j, C_i^{1,j}, C_i^{2,j}, C_i^{3,j}$)) to issue the index, anonymous public key ($C_i^{1,j}, C_i^{2,j}$) and traceability information ($C_i^{3,j}$) into the smart contract *scid*.

In our scheme, a batch of the tuple $(cid_i^j, C_i^{1,j}, C_i^{2,j})$ should be generated and submitted by TA. The vehicle V_i can use a group of the tuple to produce message/signature pair in the authentication period. If V_i runs out of all the anonymous public keys, it will request TA to replenish the new tuple.

D. Message Signing

The vehicle generates message/signature pairs in this phase. Assuming that the vehicle (e.g., V_k) wants to share the information to surrounding entities, it executes the following steps:

- 1) V_k triggers the smart contract sid and invokes $\text{Get}(cid_k^j)$ algorithm to obtain the anonymous public key $(C_k^{1,j}, C_k^{2,j})$.
- 2) V_k invokes the proof generation algorithm of SoK (in section IV-A) to compute the signature of the message msg_k , that is, $(msg_k, \sigma_k, t_k) \leftarrow \text{GenProof}(sk_k, x_k, msg_k, t_k)$, where $x_k = (C_k^{1,j}, C_k^{2,j})$, t_k is a timestamp.
- 3) Finally, the message/signature pair $(cid_k^j, msg_k, \sigma_k, t_k)$ will be sent to nearby vehicles or RSUs.

E. Message Verification

In this phase, the verifier (a vehicle or an RSU) can check the validity of the received message. If the signature is valid, the verifier accepts the message and performs further actions (e.g., changing route). The verification process of the message $(cid_k^j, msg_k, \sigma_k, t_k)$ is executed below.

- 1) The verifier checks the freshness of the timestamp t_k . If it is not fresh, the verifier rejects the message.
- 2) The verifier triggers the smart contract sid via invoking $\text{Get}(cid_k^j)$ to retrieve $(C_k^{1,j}, C_k^{2,j})$.
- 3) The verifier invokes the single proof verification algorithm of SoK (in section IV-A) **VerifyProof** to check the validity of $(x_k, msg_k, \sigma_k, t_k)$, where $x_k = (C_k^{1,j}, C_k^{2,j})$. If it is invalid, the verifier rejects the message; otherwise, it accepts the traffic-related information from V_k .

In addition, our proposed scheme can also achieve batch proof verification of multiple messages to improve efficiency. When receiving n message/signature pairs $(msg_i, \sigma_i, t_i, cid_i^j)_{i=1}^n$ from different vehicles, the verifier first triggers $\text{Get}(cid_i^j)$ from $scid$ to obtain $(C_k^{1,j}, C_k^{2,j})_{i=1}^n$. Then, it invokes the batch verification algorithm **BVerifyProof** ($\{x_i, msg_i, \sigma_i, t_i\}_{i=1}^n$) to check the validity of messages. If they are valid, the verifier accepts these message; otherwise, the verifier rejects them.

F. Track and Revocation

In our scheme, only TA can trace the malicious attacker's real identity (i.e., long-term public key). TA will proceed with the following steps.

- 1) TA first triggers $\text{Trace}(cid_k^j)$ to obtain traceable information $C_k^{1,j}$ and $C_k^{3,j}$.
- 2) Using its private key, TA can decrypt V_k 's long-term public key $PK_k = C_k^{3,j} - mskC_k^{1,j}$.
- 3) TA can retrieve V_k 's identity related to its public key.

Moreover, the anonymous public key can be revoked by TA in the following two cases.

- 1) Upon detecting the malicious behavior, TA first traces the real identity of vehicle (via Trace algorithm). Then, TA retrieves the corresponding index cid^j and revokes them via triggering $\text{Revoke}(cid_k^j)$.
- 2) If the vehicle V_k decides to unsubscribe and leave the system, it can send a revocation request with a signature to TA. Then, TA checks the signature. If it is valid, TA retrieves the corresponding cid_k^j and revokes them (via Revoke algorithm).

VI. SECURITY ANALYSIS

In this section, we demonstrate that our proposed scheme can meet the aforementioned security requirements. Specially, our EBCPPA mainly uses the SoK scheme to prevent any unauthorized traffic-related data exchange. Therefore, we first prove that our proposed SoK scheme can satisfy the security properties of the NIZK argument, including completeness, soundness, and zero-knowledge. After that, we analyze other security features of EBCPPA.

Theorem 1: If the ECDLP assumption holds, the proposed SoK scheme (see section IV-A) is a NIZK argument in the random oracle model.

Proof: We proof the security of the proposed SoK scheme in terms of *Completeness*, *Soundness* and *Zero-knowledge*.

Completeness: The completeness follows by single message verification and multiple message verification. The first case is between **GenProof** and **VerifyProof** referring a single message/signature pair generation and verification. The second case is that a batch of message/signature pairs can be checked by **BVerifyProof** simultaneously. The completeness property of the scheme can be proven as follows.

For a valid single message/signature pair $(cid_i^j, C_i^{1,j}, C_i^{2,j}, msg_i, \sigma_i, t_i)$ where $\sigma_i = \{z_i, R_i\}$, we first calculate $e_i = H(P, C_i^{1,j}, C_i^{2,j}, R_i, msg_i, t_i)$ and then the following equation always holds.

$$\begin{aligned}
 R_i &= z_i (C_i^{1,j} + P) + e_i C_i^{2,j} \\
 &= (r_i - e_i sk_i) (C_i^{1,j} + P) + e_i (u_j PK_i + PK_i) \\
 &= r_i C_i^{1,j} - e_i u_j sk_i P + r_i P - e_i sk_i P \\
 &\quad + e_i u_j sk_i P + e_i sk_i P \\
 &= r_i C_i^{1,j} + r_i P \\
 &= R_i
 \end{aligned}$$

Towards multiple message/signature pairs $(msg_i, C_i^{1,j}, C_i^{2,j}, \sigma_i, t_i, cid_i^j)_{i=1}^n$, we compute e_i , where $i \in [1, \dots, n]$, respectively. With results of e_i , the following equation always holds:

$$\begin{aligned}
 \sum_{i=1}^n v_i R_i &= \sum_{i=1}^n v_i z_i C_i^{1,j} + \left(\sum_{i=1}^n v_i z_i \right) P + \sum_{i=1}^n v_i e_i C_i^{2,j} \\
 &= \sum_{i=1}^n v_i r_i C_i^{1,j} - \left(\sum_{i=1}^n v_i e_i u_i sk_i \right) P \\
 &\quad + \left(\sum_{i=1}^n v_i r_i \right) P - \left(\sum_{i=1}^n v_i e_i sk_i \right) P \\
 &\quad + \left(\sum_{i=1}^n v_i e_i u_i sk_i \right) P + \left(\sum_{i=1}^n v_i e_i sk_i \right) P \\
 &= \sum_{i=1}^n v_i (r_i C_i^{1,j} + r_i P)
 \end{aligned}$$

Algorithm 2: Simulator SIM.**Require:** a security parameter λ_i and a generator P .**Ensure:** the message/signature pair (msg_i, σ_i) and timestamp t_i 1: Given an instance $x_i = (C_i^{1,j}, C_i^{2,j})$ to be proved.2: Randomly select $e_i^*, z_i^* \in Z_q^*$.3: Compute $R_i^* = z_i^*(C_i^{1,j} + P) + e_i^*C_i^{2,j}$.4: Set $\mathcal{O}_H = (P, C_i^{1,j}, C_i^{2,j}, R_i^*, msg_i, t_i) = e_i^*$.**return** $(msg_i, \sigma_i = (z_i^*, R_i^*), t_i)$;

$$= \sum_{i=1}^n v_i R_i$$

Soundness: On the basis of ECDLP assumption, we prove that the proposed scheme satisfies soundness under the random oracle model.

Assuming a PPT prover P^* can generate a valid instance and a message/signature pair $(x_i, msg_i, \sigma_i, t_i)$, where $x_i = (C_i^{1,j}, C_i^{2,j})$, $\sigma_i = \{z_i, R_i\}$. There exists a knowledge extractor M . M rewinds the prover P^* to the oracle query $e_i = H(P, C_i^{1,j}, C_i^{2,j}, R_i, msg_i, t_i)$. M regenerates the random oracle H such that $e_i' = H(P, C_i^{1,j}, C_i^{2,j}, R_i, msg_i, t_i)$ and continues the signing steps. After that, P^* obtains another accepted message signature pairs $(x_i, msg_i, \sigma_i', t_i)$, where $\sigma_i' = \{z_i', R_i\}$ in expected polynomial time. By combining independent equations $z_i = r_i - e_i sk_i \pmod{q}$, $z_i' = r_i - e_i' sk_i \pmod{q}$, we could get:

$$sk_i = (z_i - z_i')(e_i' - e_i)^{-1} \pmod{q}$$

Regarding P^* as a subroutine, we can break the ECDL instance with a non-negligible probability. Therefore, our proposed scheme satisfies the requirement of soundness.

Zero-knowledge: With the Fiat-Shamir heuristic method, we prove the zero-knowledge by constructing a simulator SIM which can simulate the interaction with any verifiers (see Algorithm 2).

Moreover, our scheme can also satisfy the aforementioned necessary security requirements in VANETs.

- 1) **Message Authentication/Integrity:** Since our proposed SoK scheme provides completeness and soundness, any attacker cannot forge a valid signature meeting $R_i = z_i(C_i^{1,j} + P) + e_i C_i^{2,j}$ without long-term private key. Therefore, the receiver ensures the authenticity of message sender by checking whether **VerifyProof** $(x_i, msg_i, \sigma_i, t_i)$ holds or not.
- 2) **Identity Anonymity:** In our scheme, while the real identity is cryptographically hidden from the public view, any attackers cannot extract it. The real identities of the vehicles are not transmitted during the public interaction. We use the blinded public key to achieve authentication.
- 3) **Unlinkability:** In the resignation phase, TA generates random numbers u_j to compute a batch of indexes cid_i^j of a vehicle V_i . Since each cid_i^j is used within a period time, no attacker can trace the behavior of V_i . Thus, our proposal can provide unlinkability.

- 4) **Non-Repudiation:** In our proposal, the sender cannot deny a message/signature pair that has been generated previously. The private key sk_i is a witness in the proposed SoK scheme, so only the witness owner can generate a valid signature. Supporting that the sender attempts to deny a valid message, TA could trace the real identity RID_i through $C_i^{1,j}$ and $C_i^{2,j}$. Therefore, no entity can deny the validity of the signature.
- 5) **Conditional Traceability:** Since the proposed scheme provides anonymity and unlinkability, no attacker can trace the identity of V_i . TA can obtain the ciphertext of the long-term public key via triggering **Trace**. In addition, only TA can decrypt the ciphertext with its private key msk , and then retrieve the real identity.
- 6) **Resistant against other common attacks:** Our proposal can also resist against the following common attacks.
 - **Impersonation attack:** To launch an impersonation attack, the attacker should obtain $(C_i^{1,j}, C_i^{2,j})$ from the smart contract and forge (R_i, z_i) satisfying the following equations: $R_i = z_i(C_i^{1,j} + P) + e_i C_i^{2,j}$. However, according to Theorem 1, PPT attackers cannot generate the valid pair without private key, because none of them can solve the ECDLP. Thus, our scheme can successfully resist impersonation attacks.
 - **Modification attack:** According to the above completeness and soundness analysis about our proposal, the attacker cannot forge a valid signature without the knowledge of entity's private key. Moreover, any modification of the message/signature pair could be found by checking whether the equation $R_i = z_i(C_i^{1,j} + P) + e_i C_i^{2,j}$ holds. Therefore, our proposed scheme can withstand modification attack.
 - **Man-in-the-middle attack:** Due to the analysis of message authentication, no attacker can forge valid signatures without the knowledge of private key. Also, our EBCPPA can achieve mutual authentication among participants. Hence, our proposal owns the ability to withstand man-in-the-middle attack.
 - **Replay attack:** We use a timestamp t_i and random numbers in our scheme. Upon receiving the message/signature pair, the verifier can detect the playback of messages by checking the freshness.
 - **Key escrow problem:** ID-based solutions exist key escrow problem, that is, vehicles' secret keys will be leaked once the authority (KGC) is compromised. In our scheme, the vehicle generates its key pair independently and only sends anonymous public key to authority (TA). Even if TA is compromised, TA cannot recover the secret key of vehicle from the public key. Therefore, vehicles' secret keys cannot be leaked.
 - **Stolen verifier table attacks:** There is no verifier table in our scheme. The privacy-preserving authentication only requires the vehicle to generate long-term key pair that is saved securely, and then submit anonymous public keys to blockchain. No verifier table maintained by TA, because any verifier can retrieve them from blockchain.

TABLE II
SMART CONTRACT GAS COST IN OUR PROPOSAL

Function	Gas	Actual cost	USD
Deploy	547516	0.001095032	1.7849
Submit	162658	0.000325316	0.5303
Get	0	0.000000000	0.0000
Trace	0	0.000000000	0.0000
Revoke	44372	0.000088744	0.1447

- **Birthday collision resilience:** In our design, we utilize Ethereum or other blockchain which support smart contract. These blockchain networks employ secure hash functions (e.g., SHA256) to generate the block hash. Due to the collision-resistant of secure hash functions, our scheme can avoid birthday collision attack.

VII. PERFORMANCE EVALUATION

To instantiate our proposed protocol, we implement EBCPPA scheme via the prototype of blockchain and VANETs simulation. In this section, we provide the details of the environment setting and compare our proposal with other recent approaches. The experimental performance validates its reliability and effectiveness.

A. Transaction Cost

We used the Ethereum test network (Rinkeby) to deploy our designed smart contract, where Rinkeby allows developer to test the system before deployment. Specifically, our employed configurations were a compiler (0.4.23+commit.124ca40 d), language (Solidity), EVM version (compiler default) and Environment (JavaScript VM London). After compiling our designed smart contract code (Algorithm 1), we deployed it into the selected Remix to evaluate the gas cost of each function.

On Ethereum, transaction (e.g., Submit, Revoke) requires gas which is an infrastructure cost. The external view function (e.g., Get and Trace) in our proposal is gas free. To represent gas into the monetary cost, we consider an exchange rate of 1 Ether = 1630 USD. In our configuration, the Gas Limit is set as 10,00,00 *gas* and each gas is worth 2 GWei (1 GWei = 10^{-9} Ether). On the basis of these baselines, we evaluate the gas cost in our system (i.e. Submit, Get, Trace, Revoke).

From the gas cost record shown in Table II, the maximum gas cost is the deployment of smart contract with 1.7849 USD. Fortunately, the smart contract deployment is executed only once, and the Get, Trace operation are gas free. Although other operations will be triggered repeatedly, the gas consumption of Submit and Revoke is about 0.5303 USD and 0.1447 USD respectively. Therefore, the gas cost of our proposal can be accepted in practical applications.

B. Authentication Efficiency

We further analyze the communication and computation costs of our proposal and those of other recent related BCPA schemes ([12], [13], [29]) for VANETs. We conducted our experiment

TABLE III
NOTATIONS, EXECUTION TIME AND DATA SIZE

Notation	Description	Time/Size
T_{ga}	Time of a point addition on $\mathbb{G}$	0.0811 ms
T_{gm}	Time of a scalar multiplication on $\mathbb{G}$	8.8517 ms
T_{ze}	Time of a modular exponentiation on $\mathbb{Z}_q^*$	2.7072 ms
T_{zi}	Time of a modular inversion on $\mathbb{Z}_q^*$	0.0471 ms
T_{zm}	Time of a modular multiplication on $\mathbb{Z}_q^*$	0.0108 ms
T_h	Time of a general hash function	0.0006 ms
T_{AE}	Time of an ABE encryption	390.87 ms
T_{AD}	Time of an ABE decryption	77.02 ms
$ \mathbb{G} $	Size of an element in group $\mathbb{G}$	64 bytes
$ \mathbb{Z}_q^* $	Size of an element in field $\mathbb{Z}_q^*$	32 bytes
$ ts $	Size of a timestamp	8 bytes
$ msg $	Size of a message	32 bytes
$ S_h $	Size of a transaction hash in Ethereum	32 bytes
$ S_{ABE} $	Size of an ABE ciphertext	128n+64 bytes

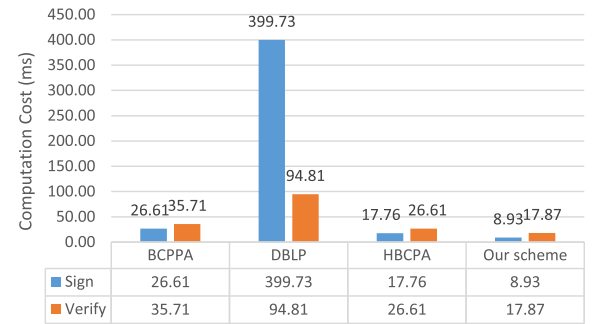


Fig. 3. Computation cost comparison (ms).

on a laptop with the Miracle V7.0 (a cryptographic library). Our timing benchmarks were on Inter (R) Core (TM) i5-4210 U 4 GB RAM at 1.7 GHz. We first evaluated the time cost of time-consuming cryptographic operations. Each operation was run 1000 times to calculate the average time. Here, we list the time cost of operations and the size of transmitted messages (see Table III). Then, we obtained the runtime of our proposed scheme and other related solutions [12], [13], [29], in terms of signature generation, single message verification and batch verification.

From the comparative results showed in Table IV and Fig. 3, we can discover that EBCPPA has superior performance than other recent solutions [12], [13], [29]. It should be noted that the proposal in [13] can achieve addition properties (such as access control), although the performance is not practical enough. The computation time of signing, verification and tracing in EBCPPA is with 8.9334 ms, 17.8662 ms and 8.9328 ms. Compared to these in [12], [13], [29], while HBCPA [29] has the best performance in real identity tracing, our scheme can save about at least 49.71%, 32.84%, in terms of signing and verification. EBCPPA avoids the use of bilinear pair operations and reduces the number of time-consuming operations. In addition, the comparison of communication overhead is shown in Table IV, which demonstrates that our scheme requires the least bandwidth (168 bytes) than that of other schemes. Therefore, our proposal is more

TABLE IV
COMPARISON OF COMPUTATION COST (MS) AND COMMUNICATION OVERHEAD (BYTES)

Item	BCPPA [12]	DBLP [13]	HBCPA [29]	EBCPPA
Message Sign.(ms)	$3T_{gm} + T_{zi} + T_{zm} + T_h = 26.6136$	$T_{AE} + T_{gm} + T_{zm} + T_h = 399.7331$	$2T_{gm} + T_{zi} + T_{zm} + T_h = 17.7619$	$T_{ga} + T_{gm} + T_h = 8.9334$
1-Message Verif.(ms)	$4T_{gm} + 2T_{ga} + 2T_{zi} + 4T_{zm} + T_h = 35.707$	$T_{AD} + 2T_{gm} + T_{ga} + T_h = 94.8051$	$3T_{gm} + T_{zi} + T_{zm} + T_h = 26.613$	$2T_{ga} + 2T_{gm} + T_h = 17.8662$
n-Message Verif.(ms)	$4nT_{gm} + 2nT_{ga} + 2nT_{zi} + 4nT_{zm} + nT_h = 35.707n$	$nT_{AD} + 2nT_{gm} + nT_{ga} + nT_h = 94.8051n$	$3nT_{gm} + nT_{zi} + nT_{zm} + nT_h = 26.613n$	$(2n)T_{ga} + (2n)T_{gm} + nT_h = 17.8662n$
Trace(ms)	$k(T_{gm} + T_h) = 8.8523k$	$T_{AD} + T_{gm} + T_{zm} = 85.8831$	$T_{gm} + T_h = 8.8523$	$T_{gm} + T_{ga} = 8.9328$
Overhead(byte)	$3 S_h + 4 Z_q^* + ts + msg = 264$	$ S_{ABE} + G + Z_q^* + msg + ts = 128n + 200$	$2 G + 2 Z_q^* + msg + ts = 232$	$ G + 2 Z_q^* + msg + ts = 168$

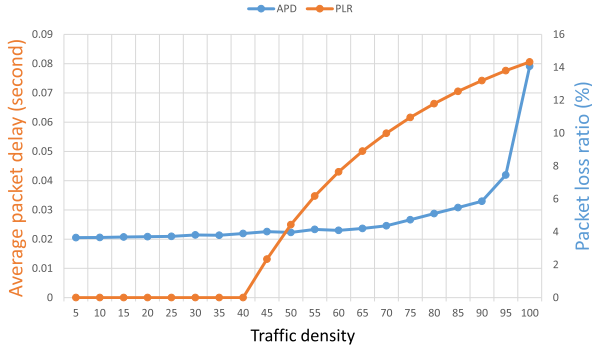


Fig. 4. APD and PLR with different traffic densities.

suitable than [12], [13], [29] in terms of computation cost and communication overhead.

C. Delay and Loss Rate

To further demonstrate the feasibility, this section evaluates the average message transmission delay and loss rate in the VANETs environment (via VanetMobSim¹ and NS-2² virtual machine). Our experiential simulations were performed on a PC running Windows 10 with 11th Gen Inter (R) Core (TM) i7-1165G7 CPU and 16 GB RAM 2.8 GHz. In the simulator scenario, we used a 1.0 * 1.0 km² virtual map with 4 RSUs equipped. The vehicles are set to communicate with each other in the range of 300 m with 6 Mbps and their speeds were ranging from 5 m/s to 70 m/s. The length of a transmitted message was 168 bytes.

We conducted two types of simulation in VanetMobSim and NS-2. In the first simulation, the speed of vehicles was controlled in a fixed range (30 m/s to 40 m/s) and the number of vehicles (density) changed from 5 to 100. The result of this experiment is shown in Fig. 4, from which we can discover that the APD changes a little before the density is added to 75, but with the growth of density the APD increases fast. Similarly, there is no data loss if the density is less than 40. When the density is more than 40, the PLR keeps increasing as the density rises.

On the basis of the former simulation, the latter experiment is for analyzing the impact of vehicles' speed on APD and PLR.

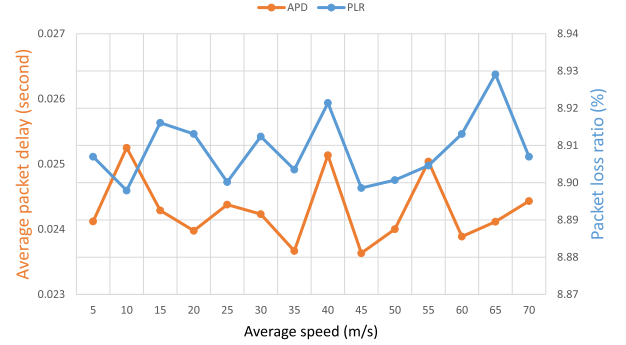


Fig. 5. APD and PLR with different average speeds.

In Fig. 5, with the increase of average speed, the APD and PLR change a little (less than 0.02 ms and 0.03%). We can conclude that the speed of vehicle has slight influence on the APD and PLR.

VIII. CONCLUSION

In this work, we proposed a more efficient blockchain-based conditional privacy-preserving authentication (EBCPPA) protocol for VANETs. We analyzed the current security requirements of practical applications and designed two building blocks, namely: SoK scheme and smart contract. In addition, we proved the security of our SoK scheme and informally discussed other security features of EBCPPA. Finally, we conducted experimental evaluations in the Rinkeby and VAENTs simulator. The results demonstrated that our scheme achieves a reduction of computation cost at least 49.71% and 32.84% in terms of signing and verification. The communication cost reduces at least 27.59% than other related schemes. Moreover, the analyses of gas cost, message delay and loss ratio also support the feasibility of our scheme. Therefore, due to its significantly improved performance, the EBCPPA scheme is more suitable to be applied in VANETs.

While we experimentally evaluate our EBCPPA in simulation environment, it is still indeterminacy in the real-world system. Further studies can be focused on implementation and optimization in practical VANETs. In addition, we will continue our effort to improve the efficiency of the CPPA and the size of the SoK proof.

¹<http://vanet.eurecom.fr/>

²<https://www.isi.edu/nsnam/ns/>

REFERENCES

- [1] A. Boukerche, H. A. B. F. de Oliveira, E. F. Nakamura, and A. A. F. Loureiro, "Vehicular ad hoc networks: A new challenge for localization-based systems," *Comput. Commun.*, vol. 31, no. 12, pp. 2838–2849, Jul. 2008.
- [2] J. Shao, X. Lin, R. Lu, and C. Zuo, "A threshold anonymous authentication protocol for vanets," *IEEE Trans. Veh. Technol.*, vol. 65, no. 3, pp. 1711–1720, Mar. 2016.
- [3] D. He, Y. Zhang, and J. Chen, "Cryptanalysis and improvement of an anonymous authentication protocol for wireless access networks," *Wirel. Pers. Commun.*, vol. 74, no. 2, pp. 229–243, 2014.
- [4] M. Raya and J.-P. Hubaux, "The security of vehicular ad hoc networks," in *Proc. 3rd ACM workshop Secur. ad hoc sensor Netw.*, 2005, pp. 11–21.
- [5] Y. Sun, R. Lu, X. Lin, X. Shen, and J. Su, "An efficient pseudonymous authentication scheme with strong privacy preservation for vehicular communications," *IEEE Trans. Veh. Technol.*, vol. 59, no. 7, pp. 3589–3603, Sep. 2010.
- [6] R. Lu, X. Lin, H. Zhu, P.-H. Ho, and X. Shen, "ECPP: Efficient conditional privacy preservation protocol for secure vehicular communications," in *Proc. IEEE 27th Conf. Comput. Commun.*, 2008, pp. 1229–1237.
- [7] D. He, S. Zeadally, B. Xu, and X. Huang, "An efficient identity-based conditional privacy-preserving authentication scheme for vehicular ad hoc networks," *IEEE Trans. Inf. Forensics Secur.*, vol. 10, no. 12, pp. 2681–2691, Dec. 2015.
- [8] M. Azees, P. Vijayakumar, and L. J. Deborah, "EAAP: Efficient anonymous authentication with conditional privacy-preserving scheme for vehicular ad hoc networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 9, pp. 2467–2476, Sep. 2017.
- [9] I. Ali, Y. Chen, N. Ullah, R. Kumar, and W. He, "An efficient and provably secure ECC-based conditional privacy-preserving authentication for vehicle-to-vehicle communication in VANETs," *IEEE Trans. Veh. Technol.*, vol. 70, no. 2, pp. 1278–1291, Feb. 2021.
- [10] X. Zheng, M. Li, Y. Chen, J. Guo, M. Alam, and W. Hu, "Blockchain-based secure computation offloading in vehicular networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4073–4087, Jul. 2021.
- [11] F. Ayaz, Z. Sheng, D. Tian, Y. L. Guan, and V. C. M. Leung, "A voting blockchain based message dissemination in vehicular ad-hoc networks (VANETs)," in *Proc. IEEE Int. Conf. Commun.*, 2020, pp. 1–6.
- [12] C. Lin, D. He, X. Huang, N. Kumar, and K.-K. R. Choo, "BCPPA: A blockchain-based conditional privacy-preserving authentication protocol for vehicular ad hoc networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 12, pp. 7408–7420, Dec. 2021.
- [13] Q. Feng, D. He, S. Zeadally, and K. Liang, "BPAS: Blockchain-assisted privacy-preserving authentication system for vehicular ad hoc networks," *IEEE Trans. Ind. Informat.*, vol. 16, no. 6, pp. 4146–4155, Jun. 2020.
- [14] M. Raya and J. Hubaux, "Securing vehicular ad hoc networks," *J. Comput. Secur.*, vol. 15, no. 1, pp. 39–68, Jan. 2007.
- [15] R. Lu, X. Lin, H. Zhu, P. Ho, and X. Shen, "ECPP: Efficient conditional privacy preservation protocol for secure vehicular communications," in *Proc. 27th IEEE Int. Conf. Comput. Commun., Joint Conf. IEEE Comput. Commun. Societies*, 2008, pp. 1229–1237.
- [16] A. Wasef and X. S. Shen, "EMAP: Expedite message authentication protocol for vehicular ad hoc networks," *IEEE Trans. Mob. Comput.*, vol. 12, no. 1, pp. 78–89, Jan. 2013.
- [17] M. Asghar, R. R. M. Doss, and L. Pan, "A scalable and efficient PKI based authentication protocol for vanets," in *Proc. IEEE 28th Int. Telecommun. Netw. Appl. Conf.*, 2018, pp. 1–3.
- [18] C. Zhang, R. Lu, X. Lin, P.-H. Ho, and X. Shen, "An efficient identity-based batch verification scheme for vehicular sensor networks," in *Proc. IEEE 27th Conf. Comput. Commun.*, 2008, pp. 246–250.
- [19] S. Hornig et al., "b-SPECS: Batch verification for secure pseudonymous authentication in VANET," *IEEE Trans. Inf. Forensics Secur.*, vol. 8, no. 11, pp. 1860–1875, Nov. 2013.
- [20] J. Li, Y. Liu, Z. Zhang, B. Li, H. Liu, and J. Cheng, "Efficient id-based message authentication with enhanced privacy in wireless ad-hoc networks," in *Proc. Int. Conf. Comput., Netw. Commun.*, 2018, pp. 322–326.
- [21] H. Dai, Z. Zheng, and Y. Zhang, "Blockchain for Internet of Things: A survey," *IEEE Internet Things J.*, vol. 6, no. 5, pp. 8076–8094, Oct. 2019.
- [22] L. Li et al., "Creditcoin: A privacy-preserving blockchain-based incentive announcement network for communications of smart vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 7, pp. 2204–2220, Jul. 2018.
- [23] M. Gonzalez-Martin, M. Sepulcre, R. Molina-Masegosa, and J. Gozálviz, "Analytical models of the performance of C-V2X mode 4 vehicular communications," *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1155–1166, Feb. 2019.
- [24] B. Luo, X. Li, J. Weng, J. Guo, and J. Ma, "Blockchain enabled trust-based location privacy protection scheme in VANET," *IEEE Trans. Veh. Technol.*, vol. 69, no. 2, pp. 2034–2048, Feb. 2020.
- [25] C. Wang, J. Shen, J. Lai, and J. Liu, "B-TSCA: Blockchain assisted trust-worthiness scalable computation for V2I authentication in vanets," *IEEE Trans. Emerg. Top. Comput.*, vol. 9, no. 3, pp. 1386–1396, Jul.–Sep. 2021.
- [26] S. Rowan, M. Clear, M. Gerla, M. Huggard, and C. M. Goldrick, "Securing vehicle to vehicle communications using blockchain through visible light and acoustic side-channels," Apr. 2017, *arXiv:1704.02553*.
- [27] Z. Lu, Q. Wang, G. Qu, H. Zhang, and Z. Liu, "A blockchain-based privacy-preserving authentication scheme for vanets," *IEEE Trans. Very Large Scale Integr. Syst.*, vol. 27, no. 12, pp. 2792–2801, Dec. 2019.
- [28] D. Gabay, K. Akkaya, and M. Cebe, "Privacy-preserving authentication scheme for connected electric vehicles using blockchain and zero knowledge proofs," *IEEE Trans. Veh. Technol.*, vol. 69, no. 6, pp. 5760–5772, Jun. 2020.
- [29] X. He, X. Niu, Y. Wang, L. Xiong, Z. Jiang, and C. Gong, "A hierarchical blockchain-assisted conditional privacy-preserving authentication scheme for vehicular ad hoc networks," *Sensors*, vol. 22, no. 6, 2022, Art. no. 2299.
- [30] Y. Tsiounis and M. Yung, "On the security of elgamal based encryption," in *Int. Workshop Public Key Cryptography*, 1998, pp. 117–134.
- [31] M. Blum, P. Feldman, and S. Micali, "Non-interactive zero-knowledge and its applications (extended abstract)," in *Proc. 20th Annu. ACM Symp. Theory Comput.*, (J. Simon, ed.), 1988, pp. 103–112.
- [32] M. Chase and A. Lysyanskaya, "On signatures of knowledge," in *Proc. Annu. Int. Cryptology Conf.*, 2006, pp. 78–96.
- [33] A. Dorri, M. Steger, S. S. Kanhere, and R. Jurdak, "Blockchain: A distributed solution to automotive security and privacy," *IEEE Commun. Mag.*, vol. 55, no. 12, pp. 119–125, Dec. 2017.



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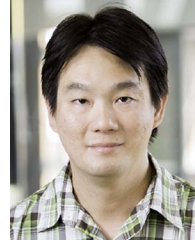
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